

METRICMIND

Agentic Semantic BI Engine

Final Project Report — Requirements-Aligned Submission

Axlero Solutions • Project 1 • August 2026

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1. Project Requirement

The source assignment defines MetricMind as an Agentic Semantic BI Engine for Enterprise Analytics and Agentic AI. The problem is trust: unrestricted Text-to-SQL can produce incorrect joins, inconsistent metrics and results that are hard to audit.

The expected flow is a natural-language business question → governed semantic layer → controlled data result → business explanation. The source example is: "Why did our European margins drop last quarter?"

- Governed semantic layer for business metrics and dimensions.
- Agent/orchestrator using semantic APIs rather than unrestricted SQL.
- Analytical warehouse/lakehouse layer.
- Conversational BI interface with charts and traceability.

2. Required Architecture

- Semantic Layer — Cube.dev or dbt; governed Revenue, Churn, Margin and dimensions.
- Agentic Orchestrator — LangChain / Llama 3; natural language to semantic API calls.
- Data Platform — Snowflake / Databricks.
- Conversational BI — Next.js + Tremor with streaming UI and ECharts.

Repository status: the implementation uses React + Vite, FastAPI + Python and MySQL. The source stack and the repository stack are therefore not the same, and this report does not present them as equivalent without proof.

3. Four-Week Plan and Acceptance Checks

Week 1 — warehouse/modeling and chat scaffolding.

Week 2 — semantic definitions and LangChain prompts.

Week 3 — governance audit and API check.

Week 4 — multi-step reasoning, dynamic visualizations, cost governance and View API/SQL.

Acceptance checks include repeated “Q3 Revenue” consistency, correct semantic API translation for European sales, a secondary cost breakdown for margin drops, suitable charts, query limits, and View SQL / View API Call transparency.

4. Implemented Repository

Frontend: React + Vite — Dashboard, Reports, Chat and Profile.

Backend: FastAPI + Python — REST APIs and authentication.

Database: MySQL — factinternetsales plus dimension tables.

Documentation: README, API, installation, testing, bug report and user guide.

Repository: ShyamVarmora/MetricMind; default branch: devops.

5. Application and API Evidence

The evidence pack contains dashboard, chart, recent-transaction, Swagger/OpenAPI, authentication and runtime screenshots. These show that the team exercised a usable application and API surface during the recorded workflow.

Authentication/profile, dashboard/KPI/chart areas and the main navigation areas were exercised.

UI presence is not treated as proof of the full semantic-layer → agent → governed API → explanation workflow.

6. Testing Record

Historical test session: 14 APIs tested; 8 passed; 6 failed.

The six failures were recorded in Home Dashboard, Reports and Analytics backend APIs. Three high-priority open bugs were documented.

The historical results are kept separate from the latest local working-tree state and are not presented as a fresh regression pass.

7. Current Working-Tree Note

On 16 August 2026, the latest local terminal output showed Uvicorn stopping with `SyntaxError: ":" expected after dictionary key` at `backend/app/main.py` line 176.

The GitHub devops branch currently contains a syntactically valid `main.py` at its latest committed revision. The clean check is therefore: discard/restore local uncommitted changes, pull devops, start backend, start frontend, then repeat the API checks.

8. Governance Requirements

No raw SQL: the agent should use governed semantic APIs.

Consistent metrics: definitions should be centralized and mathematically explicit.

Query limits: expensive exploratory queries should be bounded.

Traceability: the reasoning/data path and API/SQL trace should be visible where applicable.

These are source requirements. The available evidence does not prove full end-to-end implementation of them.

9. Axlero Execution and GitHub Requirements

Axlero SOP v1.1 requires two projects out of three across a two-month cycle, with four week-wise implementations per project. GitHub is mandatory for non-UI/UX domains.

Final review checks Weeks 3–4 and requires GitHub activity on every day of the preceding 20-day period, with no gaps.

A Work Distribution Document is required before the Month 1 Mid Review. Team-lead duties include repository/branch management, work distribution, reviews and merges.

10. Team

Shyam — Team Lead • DevOps • Integration.

Chetan — Backend • Agent orchestration.

Mansi — Data • Database • Semantic validation.

Yoshita — UI / Frontend.

Himesh — Frontend • Reassigned work.

Prashant — QA • API testing • Documentation.

11. Final Assessment

The documentation submission is complete: the report and presentation cover the source requirements, architecture, week-wise plan, repository implementation, testing record, governance targets, team responsibilities and GitHub requirements.

Proven: application foundation, authentication evidence, dashboard UI evidence, repository documentation and historical testing evidence.

Not proven: full Cube/dbt semantic-layer compliance, full LangChain/Llama 3 agentic orchestration, Snowflake/Databricks implementation, the complete multi-step root-cause workflow, end-to-end cost governance and View SQL / View API Call transparency.

The distinction is deliberate. Unsupported implementation is not claimed as completed.

12. Source Basis

Axlero Solutions — Project Document – Advanced Data Analytics, Project 1: “MetricMind”: Agentic Semantic BI Engine.

Axlero Solutions — Standard Operating Procedures for Intern Projects Version 1.1.

MetricMind Project Execution Guide (Phase 1) and Project Execution Master Plan.

Repository README/API/testing/installation/bug/user-guide documentation and the supplied evidence pack.